

EMPIRICAL MODELS OF HEIGHT INTEGRATED CONDUCTIVITIES

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Abstract. Two-dimensional distributions of the height-integrated Pedersen and Hall conductivities have been computed for latitudes poleward of 60° invariant representative of two activity levels: $0 \leq K_p \leq 3_0$ and $3_0 < K_p \leq 9+$. Average precipitating fluxes of electrons with energies of 0.15, 1.27, 9.65, and >22 keV obtained by the energetic particle detector of Isis 2 during 1971-1974 are used as input to a Rees-type computation. The assumption of equilibrium conditions and a recombination rate profile permit calculation of electron density profiles and conductivity profiles. Calculations are performed at 300 grid points, specifically 12 local times and 25 latitudes from 60° to 84° invariant latitude. The models include ionization due to galactic EUV and other background sources that produces base conductivities ($\Sigma_p \sim 0.1$, $\Sigma_H \sim 0.2$ mhos) as well as solar photon ionization through an empirical fit to Chatanika radar observations [Mehta, 1979]. Substantial modulation of the conductivities is found to result from longitudinal variation of the magnitude of the earth's magnetic field. The results show considerable spatial gradients and are expected to be of use in numerical modeling.

Introduction

In recent years, several attempts have been made to model aspects of the magnetosphere-ionosphere current system [Yasuhara et al., 1975; Harel and Wolf, 1976; Kisabeth, 1979]. In common these efforts have been based upon rather simplistic models of the ionospheric conductivities, which can lead to difficulties because conductivity gradients may be of equal importance to electric field gradients as the source of horizontal current divergence [Wallis, 1976].

We present here two dimensional models of the height-integrated Pedersen and Hall conductivities based upon average distributions of precipitating electrons observed by the energetic particle detector on the polar orbiting satellite Isis 2. Two models are computed based upon particle averages for $0 \leq K_p \leq 3_0$ and $3_0 < K_p < 9+$. These models can be used with similar average electric field distributions [e.g., Heppner, 1977] to infer three-dimensional current systems or perhaps in the self-consistent models such as those of Vasyliunas [1972] or Harel et al. [1978].

The conductivities are calculated from standard (classical) formulas and computed electron density profiles. The electron density profiles are computed from ionization rate profiles calculated by the method of Rees [1963, 1964]. Details of these calculations are outlined in the next section together with statements of other assumptions. Plots of the resulting conductivity element distributions are then presented and briefly discussed.

Method

Computation of the altitude profile of the (classical) conductivity tensor elements is a straightforward process given an electron density profile. One requires only a knowledge of magnitude of the magnetic field strength and the collision-frequency profiles. Such computations have been performed extensively in the past [for example, Rowe and Mathews, 1973; Brekke et al., 1974; Vanyan and Osipova, 1975]. A model atmosphere is assumed, the collision frequencies are computed, and the conductivity elements then result from the application of well-known formula [Hanson, 1965; Whitten and Poppoff, 1971]. As these procedures, which are used in this work, are well established in the literature, they are not discussed in detail here but are reviewed briefly in Appendix 1.

Unlike the references mentioned above, radar observations of electron density profiles were not used; instead, we calculate two-dimensional distributions of the conductivities based on observed distributions of precipitating electrons. This method allows the computation to be made over a greater latitude range than can be observed by a single radar. However, electron density profiles must themselves be computed.

Rees [1963, 1964] presented a computational method whereby the ion pair production profile may be calculated for arbitrary electron energy and certain pitch angle distributions based upon laboratory measurements of electron beam energy deposition as a function of partial range. This method has been employed extensively in the past [e.g., Rees and Jones, 1973; Jones, 1975] for the computation of ion pair production profiles and other quantities derivable from the energy deposition profiles. We employ the Rees methodology here in a time static case for the calculation of ion pair production rate profiles. Details of this calculation are described in Appendix 1.

From the ionization rate profile computed by the Rees method we have computed an electron density profile under the assumption of equilibrium. Effective recombination rates of Rowe [1972] below 90 km, and Watt et al. [1974], 90 to 200 km, were used. Because a small contribution to the height-integrated Pedersen conductivity can result from ionization above 200 km, the Watt et al. values have been smoothly extrapolated above that altitude. Since Watt et al. values of the effective recombination coefficient have been derived from auroral zone radar measurements, they have been used in the hope that they reasonably reflect average conditions of the auroral ionosphere.

Transport effects, which become increasingly important above 200 km due to long lifetimes, smear out the ionization and reduce the contribution to the Pedersen height-integrated conductivity. In order to estimate the contribution of the lower F region to the conductivities, we integrate to 220 km by using the extrapolation of the re-

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combination rates. Until transport terms (and in particular the meridional wind field) are fully understood, no better estimate of the conductivities can be made by this technique. Properly stated then we are computing the classical height-integrated E region conductivities and are estimating the contribution of the F region.

The particle observations employed in this work are average distributions of precipitating electrons measured by the energetic particle detector instrument furnished by the National Research Council of Canada for the polar orbiting Isis 2 spacecraft. Detector characteristics have been presented by Venkatarangan et al. [1975]. Contour maps of some of these average particle distributions were published by McDiarmid et al. [1975]. Their paper describes the procedures employed to obtain the average distributions in bins 1° of (invariant) (Λ) latitude by 2 hours of local time (MLT). Numeric values upon which the contour maps were based are used in the present work and are presented in Tables A1 and A2 of Appendix 2. However, the present work differs from McDiarmid et al. in three respects: (1) A larger data set is employed such that ~ 650 data points from ~ 150 passes were averaged in each Λ latitude-MLT bin, (2) a smaller unpublished data set for $3 < K_p < 9$ derived in the same way is employed, and (3) a correction has been applied to the average fluxes for detector counting efficiency based on an inter-comparison of data with that obtained by the University of Texas at Dallas soft particle spectrometer also onboard Isis 2.

Average loss cone intensities for every latitude/time bin were available at five energies: 0.15, 1.27, and 9.65 keV (differential measures) and >22 and >210 keV (integral measures). The >210 keV measurements are not utilized in this study. For each bin a power law, $I = I_0 E^{-\alpha}$, was fitted to successive pairs of intensity data at adjacent energies. (It is possible to fit a power law to the 9.65 differential and >22 integral points by successive approximations.) Power law constants for the .15 to 1.27, 1.27 to 9.65, and >9.65 keV energy bands were then employed to evaluate precipitating electron fluxes at 33 logarithmically spaced energies between 0.15 and 71 keV (the 9.65 to >22 keV fit was extrapolated to 71 keV). These were used as input spectra for the Rees method computations. It is assumed that these average fluxes are isotropic in pitch angle.

In order to be as realistic as possible the model includes both solar EUV and background (non-auroral) particle effects. Conductivities produced by these sources cannot simply be added to the conductivities generated by particle precipitation. Instead, it is necessary to add the ionization rates at each altitude and compute the conductivities based on these total ionization rates. At low latitudes, galactic EUV and cosmic radiation, scattered EUV and weak diffusion from the radiation belts all contribute to ionospheric ion production. These same sources, excepting the radiation belt precipitation, must be present at polar cap latitudes as well. To include these sources, we have constructed a background electron density profile $N_B(z)$, and from that a background ionization rate profile $P_B(z) = \alpha(z)N_B^2(z)$, utilizing the assumed recombination rate profile described above. The ionization rate profile was assumed to be representative at all local times and latitudes.

Following Harris and Tohmatsu [1972] the assumed nighttime background electron density N_B has an E region peak at 110 km of $2.5 \times 10^9 \text{ m}^{-3}$ and densities of $1.4 \times 10^9 \text{ m}^{-3}$ at 100 and 120 km. Below 100 km the background Ne profile decreases exponentially to a density of $2.6 \times 10^8 \text{ m}^{-3}$ at 76 km and more steeply below that altitude to match the results of Rowe [1972]. The E-F region minimum density was $1.25 \times 10^3 \text{ m}^{-3}$ at 140 km. Above 140 km the Ne profile was assumed to increase linearly with altitude to a density of $4 \times 10^9 \text{ m}^{-3}$ at 220 km. This latter value was suggested by Isis sounder measurements as being a representative minimum F region density (J. H. Whitteker, private communication, 1976).

Conductivities produced by solar illumination are modeled by an approximation, based upon empirical data, in order to reduce computing time. In essence, a magnetically quiet daytime ionization rate profile is scaled by a solar zenith angle dependent (but altitude independent) function. This scaled production profile is then added to those of the background and auroral particles. Mehta [1979] fitted the height-integrated conductivities Σ_H and Σ_P derived from Chatanika electron density measurements with a linear function of solar zenith angle χ . He employed data from a magnetically quiet day that were readily edited to remove the effects of one small substorm. Mehta's results are $\Sigma_P = 12.579 - 0.122 \chi$ (mhos) and $\Sigma_H = 24.421 - 0.239 \chi$ (mhos), where χ is in degrees, valid in the range $45 \leq \chi < 95^\circ$.

We employ a quiet day electron density profile $N_S(z)$ obtained from Chatanika observations (Figure 1 of Brekke et al. [1974]) to derive a quiet day ionization rate profile $P_S(z) = \alpha(z) N_S^2(z)$ via the recombination rate profile $\alpha(z)$ described previously. Next $P_S(z)$ is scaled by a factor β^2 ; the equilibrium electron density $N_S'(z) = (\beta^2 P_S(z) / \alpha(z))^{1/2}$ and the height-integrated conductivities are calculated for several values of β . These height integrations are carried out from 90 to 170 km, as done by the Utah group, and using an atmospheric model appropriate to Chatanika at the time of Mehta's source observations. Comparing these height-integrated conductivities $\Sigma_H(\beta)$ and $\Sigma_P(\beta)$ with Mehta's fits, we find that $\beta = 3.92 - 0.0382 \chi$ reproduces Mehta's results within 6.2% for $45^\circ \leq \chi \leq 95^\circ$, which is judged to be satisfactory. We are thus able to compute $P_S(z, \chi) = P_S(z) \beta^2(\chi)$ as necessary.

In practice, the solar zenith angle χ is computed for each grid point under the assumption that the invariant latitude of the grid point is equal to its centered dipole latitude. With this assumption the invariant latitude and magnetic local time can be used together with universal time to compute the geographic coordinates of each grid point and the solar zenith angle may be computed in the normal way.

There are several small errors that result from this approximate method. The solar zenith angle may be in error by up to 5° due to the noncoincidence of centered dipole and invariant latitudes. Over North America, however, this error is generally less than 1° . Additionally, there should be two scaling functions, $\beta(\chi)$, one for each conductivity element, but we employ the median of these two in order to reduce computational costs. Over the range of applicability of Mehta's fit ($45 \leq \chi \leq 95^\circ$) this error amounts to 1% in Σ_H and Σ_P at $\chi = 45^\circ$ and 6.2% at $\chi = 95^\circ$. For the present pur-

poses, these errors are certainly satisfactory (see later comments on probable errors). Obviously the approximate method would, if the atmospheric model remained static, yield a fixed ratio Σ_H/Σ_P independent of χ . However, since we are allowing the atmosphere to vary as a function of the time of year, local solar time, latitude, etc., the conductivity ratio is not fixed. The addition of these atmospheric model dependencies thus contributes an unknown but probably small uncertainty in the resulting conductivities. Finally, we employ this methodology over the range $95^\circ \leq \chi \leq 102^\circ$. Obviously, as χ exceeds 95° , Σ_H must fall off more rapidly than Σ_P owing to the loss of E region solar production, while the F region is maintained. The conductivities themselves are, however, small relative to those of moderate auroral particle input and this error is again negligible at most grid points.

In this work we have employed the MSIS model atmosphere [Hedin et al., 1977a, 1977b]. This MSIS atmosphere has no universal time or longitude variations but does possess latitude and local solar time variations which make small changes in the results. In the present case, average conditions ($A_p = 4$, $F_{10.7} = \bar{F}_{10.7} = 150$) have been assumed. The atmosphere is evaluated for each grid point.

Results

Figure 1 presents contour maps of the height-integrated Pedersen and Hall conductivities and 3914-Å emission due to electron precipitation alone for the two K_p conditions. In addition, these quantities are tabulated in Appendix 2. The solar (background) source has been turned off for these computations, although its influence is still present through the model atmosphere. These results were computed for 17 hours UT of April 1 (day 91). The basic structure of the auroral oval is discernible in each of these contour maps; i.e., the oval's diurnal latitude variation is present as well as morning sector broadening and thinning near the noon sector. While these results are a convolution over energy, all of these plots show features near 15 MLT and $75^\circ \Lambda$ corresponding to the soft precipitation enhancement noted by McDiarmid et al. [1975].

Consider first the Pedersen conductivity maps. Finite nonzero conductivities are found in the central polar cap. The 82° to $84^\circ \Lambda$ average values (all local times) for Σ_P are 0.78 mhos, $0 < K_p < 3$, and 0.32 mhos, $3 < K_p < 9$. These conductivities are principally caused by ~ 1 -keV electrons that appear here as a relatively uniform drizzle precipitation. In reality they represent discrete forms ('sun-aligned' arcs, patchy forms) as well as more uniform drizzle events. Such precipitations are in general anticorrelated with magnetic activity, so the decreased conductivities for the high K_p case seem reasonable. These particles produce little 3914-Å luminosity, as can be seen from the 3914 contour maps. Not all of this conductivity is produced by particle precipitation, as about 0.1 mhos is produced by the background sources described previously.

Within auroral latitudes the peak height-integrated conductivity produced is 8.7 mhos at $66^\circ \Lambda$ and 0.3 MLT for $0 < K_p < 3$ and 12.8 mhos at $66^\circ \Lambda$ and 05 MLT when $3 < K_p < 9$. For the low K_p case the two $\Sigma_P > 7$ mhos 'islands' at ~ 0400 and 2200

MLT are separated by a shallow sea $\Sigma_P = 6$ mhos across midnight. Similarly, the two $\Sigma_P > 10$ mhos features in the high K_p case have a minimum of $\Sigma_P = 7$ mhos between them.

It is perhaps more interesting that larger conductivities occur in the noon sector ($\sim 73^\circ \Lambda$) in the low K_p case. Thus the low K_p $\Sigma_P = 3$ mhos contour is continuous across noon, whereas it is the $\Sigma_P = 2$ mhos contour which would appear continuous in the high K_p case. The average electron energy in this region is ~ 3 keV for both K_p ranges.

At the low-latitude nightside, Σ_P has not dropped to its background value (~ 0.1 mhos) although it is well down from its peak within the oval. Extrapolation of the trends for both K_p cases reveals that background level would have been achieved had two more degrees of data been available, i.e., the conductivities and luminosities would not be different from background at $58^\circ \Lambda$. Near noon, however, $\Sigma_P \sim 1$ mhos at the lowest latitudes and there is no trend of significance that would suggest that Σ_P will decay to background levels equatorward of $60^\circ \Lambda$. These conductivities are produced by ~ 3 -keV electrons and probably result from a low level drizzle from the radiation belts or possibly from a residual plasma sheet population [Wallis et al., 1979]. These low-latitude dayside particle conductivities are of little importance, however, as we shall soon demonstrate.

An almost identical description holds for the height-integrated Hall conductivity, excepting, of course, that the magnitude of Σ_H within the oval is greater. For $0 \leq K_p \leq 3$ the polar cap average (82° to $84^\circ \Lambda$) Σ_H is 0.85 mhos and the largest oval value is 12.9 mhos (at $67^\circ \Lambda$ and 05 MLT). Corresponding values for $3 < K_p < 9$ are 0.47 mhos and 27.3 mhos (at $65^\circ \Lambda$ and 05 MLT).

The 3914-Å emission contour plots are included in Figure 1 primarily as means of establishing confidence in the conductivity results. That the model 3914-Å emission contours resemble the oval so well verifies both the approach of using the average particle data and the method in detail as far as the computation of the ionization rate profiles.

We turn now to the effect of the sun. Figure 2 presents contour maps of Σ_P and Σ_H with solar and background ionization sources appropriate for 1700 UT of day 91, with the same atmospheric determining parameters used for Figure 1. At noon, the lowest latitude points have $SZA > 45^\circ$ so that the solar conductivity model should be valid everywhere throughout the plot (the $\chi = 95^\circ$ line approximately coincides with $\Sigma_P = 1$ mhos). The obvious tilt of the dayside contours is not caused by the local time variation of the model atmosphere, but rather it is due to longitudinal asymmetry in the magnitude of the earth's magnetic field, which is a parameter in the classical conductivity tensor (see Appendix 1). This effect arises as a consequence of fixing the universal time in order to fix the relationship between magnetic coordinates and geographic coordinates. Other tilts are obtained for other choices of universal time.

The dependence of conductivities upon longitude does not seem to have been discussed in the literature. Figure 3 presents the relative variation of Σ_H and Σ_P along the $\Lambda = 65^\circ$ circle in the northern hemisphere. A flux of 4-keV electrons was used in this computation and the results are normalized to 0° INV longitude. Two curves are given for each conductivity element, with and without variations in the model atmosphere. Relative va-

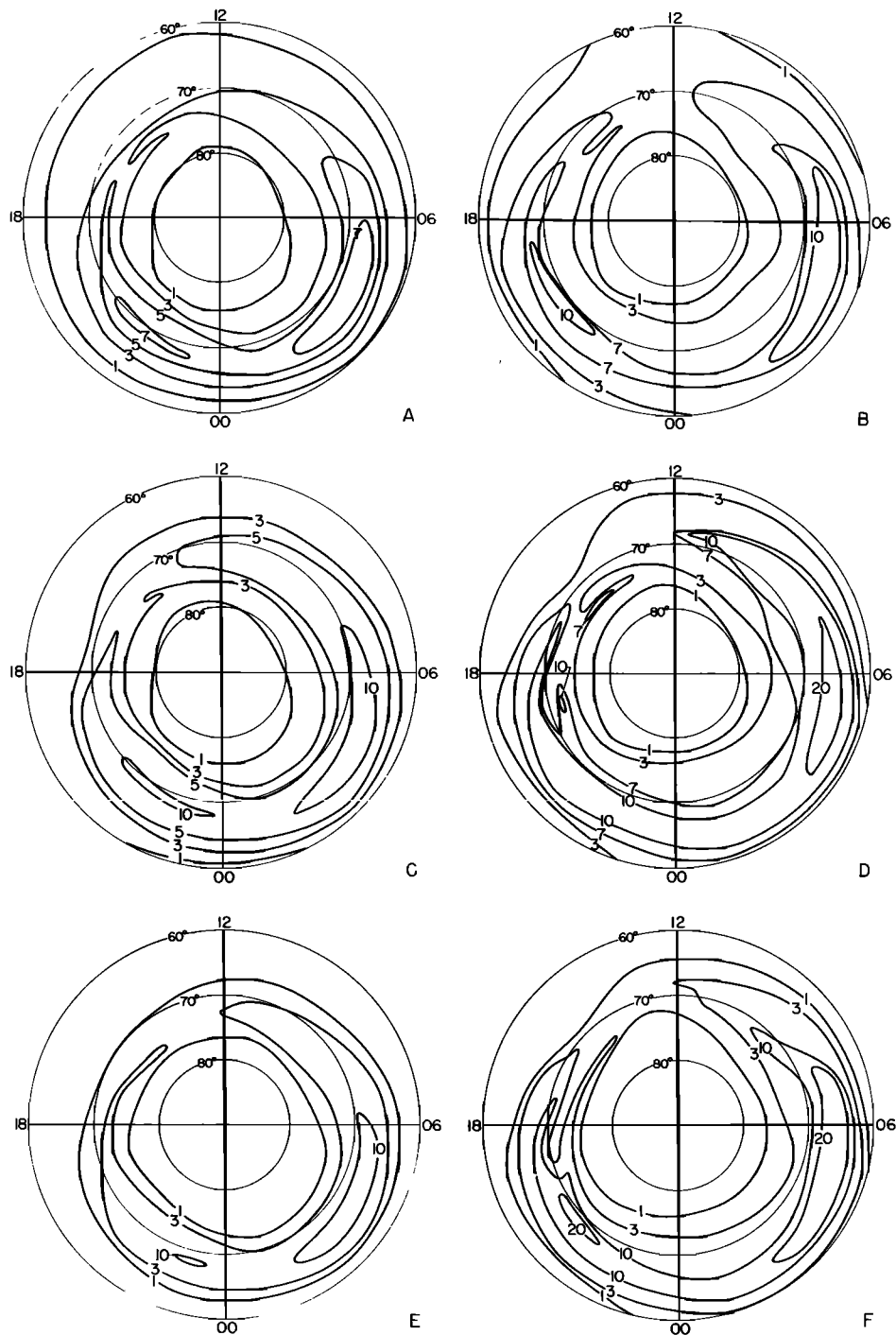


Fig. 1. (a) Pedersen height-integrated isoconductivity contours (mhos) in an invariant latitude and magnetic local time frame produced by average particle distribution determined by Isis 2 EPD and background ionization sources for $0 \leq K_p \leq 30$. (b) Pedersen isoconductivity contours (mhos) as in (a) for $K_p > 30$. (c) Hall isoconductivity contours (mhos) as in (a) for $0 \leq K_p \leq 30$. (d) Hall isoconductivity contours (mhos) as in (a) for $K_p > 30$. (e) Height-integrated 3914-Å emission rates as in (a) for $0 \leq K_p < 30$, in units of 100 R. (f) Height-integrated 3914-Å emission rates as in (a) for $K_p > 30$, in units of 100 R. The conditions specified for the computations are $A_p = 4$, $F_{10.7} = \bar{F}_{10.7} = 150$ for day 91 at 1700 UT. Coordinates are GSFC invariant latitude and centered dipole (pole location 78.5°N , -69.0°E) magnetic local time. All calculations are performed for the northern hemisphere.

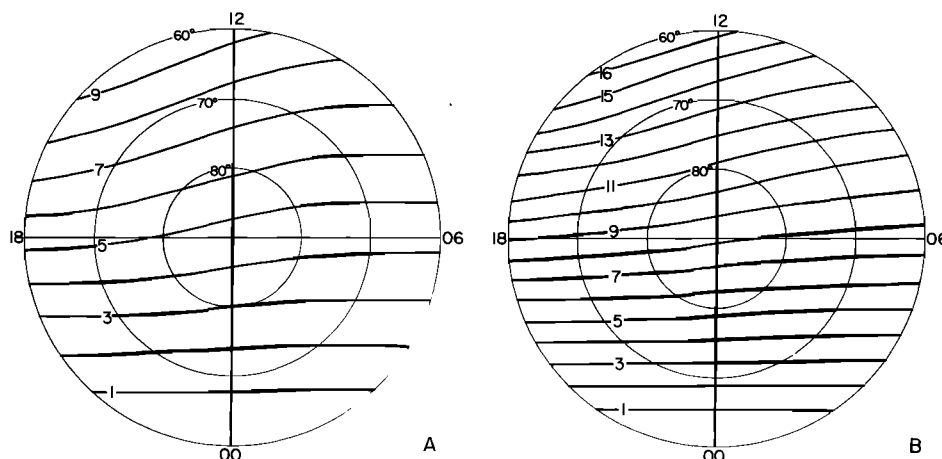


Fig. 2. (a) Pedersen height-integrated isoconductivity contours (mhos) and (b) Hall isoconductivity contours (mhos) produced by solar illumination and background appropriate for the same time and atmospheric model as Figure 1.

variation in the conductivities produced by a constant particle flux is even larger in the southern hemisphere.

Figure 4 presents contours of the two conductivity elements for both K_p cases when background, solar and particle sources are present. As expected the dayside conductivities are dominated by the sun and there the contours tend to show alignment perpendicular to the noon meridian. The nightside is still dominated by the particles. A broad pocket is formed just poleward of the midnight oval by the combined effects of the oval structure of the particles and the sunward pointing gradient of the solar produced conductivities.

Discussion

We have calculated two dimensional distributions of the height-integrated Hall and Pedersen conductivities from average particle distributions assembled for two K_p ranges. The conductivity distributions show considerable spatial structure and should be more representative of the actual conductivity distribution than have been employed in the past in numerical modelling experiments.

Two models were computed for two ranges of the K_p index, since these allow estimation of changes with increased auroral activity. Table 1 gives the mean value of K_p obtained from each data set as well as hemispheric integrals (60° to 84° Λ) over the particle distributions. These latter quantities are essentially in agreement with accepted values.

A large number of error sources occur in these calculations as virtually every step requires some approximation or relies upon measured quantities. Potentially the largest error lies in the assumption that classical (collision) conductivity formulas are appropriate. Collective plasma phenomena do occur in the conductivity generating regions of the ionosphere, e.g., radar aurora, but their influence upon conductivities is at present poorly defined. Undoubtedly the largest error in the present calculations lies with the measured particle fluxes. While everything possible has been done to remove systematic errors here, there may be as much as a factor of 2 error in the particle fluxes. There may, therefore, be a factor of $\sqrt{2}$

error in the conductivities from this source alone.

We have found, as Brekke et al. [1974] found, a small ($\sim 10\%$) dependence upon the atmospheric model expressed in terms of the exospheric temperature. The assumption of an electron temperature profile does not contribute substantial error. However, substantial error may result from assumption of equilibrium. This assumption may break down in regions of the oval where rapid temporal changes occur, most prevalently in the midnight sector, and it is difficult to estimate the probable effect upon these computations. Both the recombination rate profile and the assumed form of the ion collision frequency may also con-

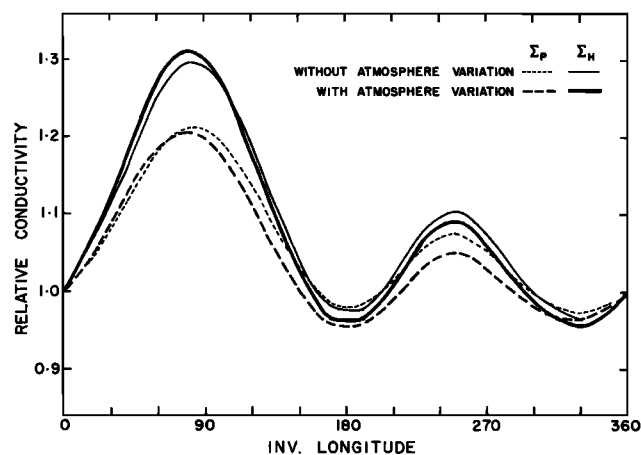


Fig. 3. Longitudinal dependence at 65° north invariant latitude of height-integrated Hall (solid line) and Pedersen (dashed line) conductivities normalized at 0° invariant longitude with (thick line) and without (thin line) model atmospheric variations. These computations are performed for a unit flux of isotropic 4 keV electrons with the model atmosphere specified by $A_p = 4$, $F_{10.7} = F_{10.7} = 150$. Solar illumination is not present. Minimum and maximum exospheric temperatures occur at 70° and 210° INV LONG, respectively. The Lockheed [Evans et al., 1969] invariant coordinates are used.

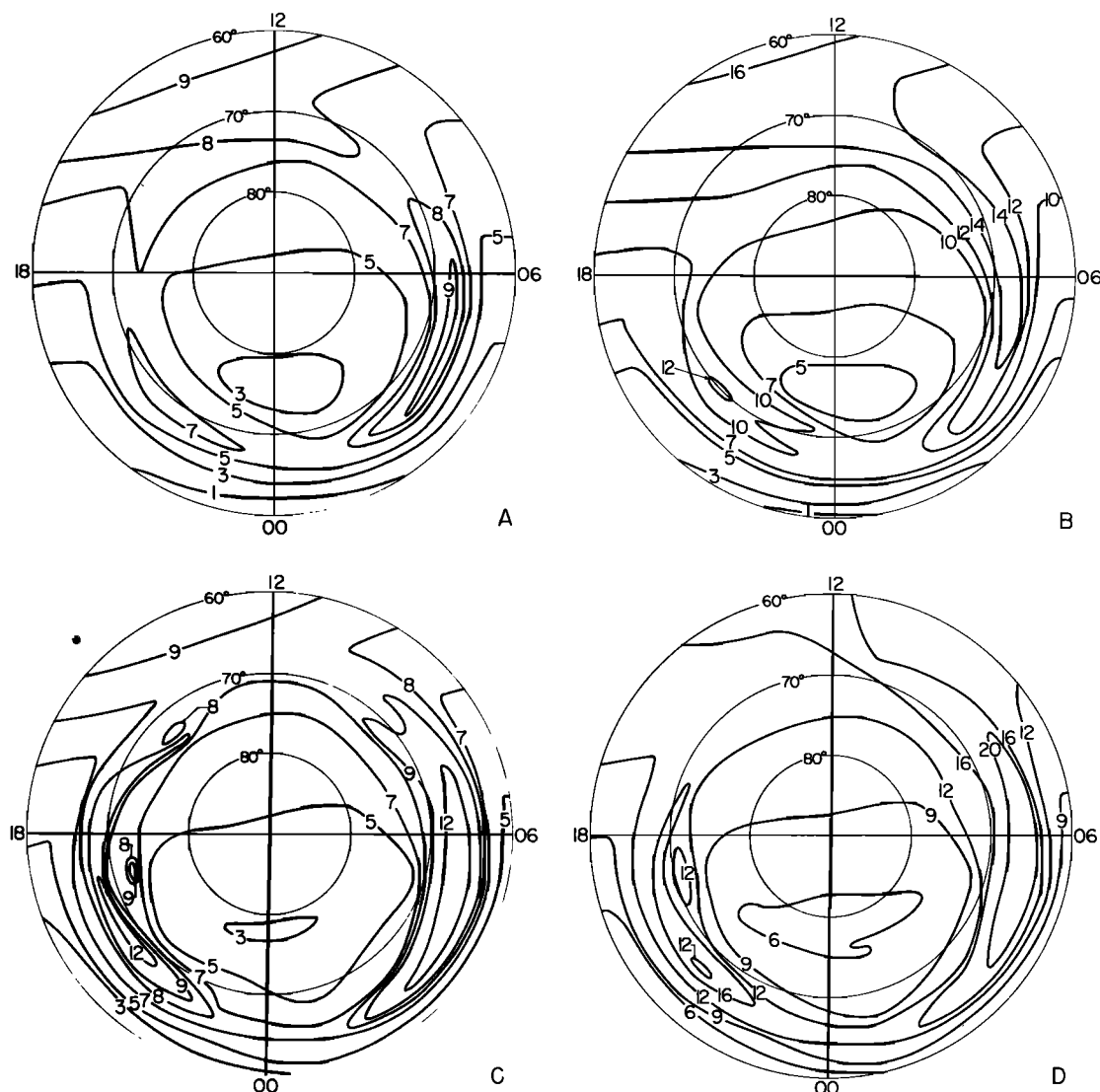


Fig. 4. Contour plots of Σ_p and Σ_H (mhos) for the two K_p cases produced by particle, background, and solar sources for 1700 UT on April 1. (a) Σ_p for $0 \leq K_p \leq 30$, (b) Σ_H for $0 \leq K_p \leq 30$, (c) Σ_p for $30 < K_p$, (d) Σ_H for $30 < K_p$.

tribute substantial uncertainty. Considering all sources of error, we estimate that the present calculations for the conductivities are probably good within a factor of 2.

Additionally, the reader should be cautious in applying these present results to individual events. The averaging process by which the 'input' particle distributions were obtained both broadens the distribution and reduces precipitation peaks. Thus particle produced conductivities can be expected to be larger and more structured in individual events.

Two other factors have been ignored in the present calculations; the production of ionization by positive ion precipitation and the modification of ionization levels in the afternoon trough. The first of these is probably generally unimportant as significant positive ion energy fluxes are observed to be rare. The second, neglected, factor cannot be modeled directly in the time static case examined here.

It is not possible here to reproduce more con-

tour plots of the height-integrated conductivities Σ_p and Σ_H which would be representative of other seasons and local times. We can, however, obtain a reasonable impression of their appearance by inspection of Figures 1, 2, and 4. Obviously, at winter solstice when the magnetic pole is directed away from the sun (say, day 355 at 0500 UT for the northern hemisphere) the conductivities will be dominated by the particles and should be basically similar to the situation shown in Figure 1. At 60° Λ and noon under these circumstances the solar zenith angle χ is nominally 95° and thus alters the noon contours only slightly at low latitudes. Similarly, as summer solstice approaches, we expect more and more of the polar region will be dominated by sunlight, and the contours will show a greater tendency to align themselves perpendicularly to the noon-midnight meridian. At summer solstice with the magnetic pole toward the sun, central polar cap conductivities are ~ 1.7 times those shown in Figure 2.

One can obtain an approximate value of the to-

TABLE 1. Hemispheric Integrals of Particle Distributions and Modeled Quantities (60° to 84° Λ)

Activity Level	K _p Range	Mean K _p	Precipitation Energy Flux erg s ⁻¹	Particle Flux, s ⁻¹	Average Energy, $\bar{E}$, keV	3914-Å Emission Photon/s	Mean Luminous Efficiency R erg ⁻¹ cm ² s
Low	0 to 3 ₀	1+	6.54 × 10 ¹⁶	2.08 × 10 ²⁵	1.96	2.65 × 10 ²⁵	558
Moderate	3+ to 9	4 ₀	1.19 × 10 ¹⁷	2.91 × 10 ²⁵	2.54	4.94 × 10 ²⁵	579

tal conductivity Σ_{iT} from the conductivities produced by the sun Σ_{iS} and particles Σ_{iP} by $\Sigma_{iT} = \{\Sigma_{iP}^2 + \Sigma_{iS}^2\}^{1/2}$. Sample calculations indicate this expression to be in error by ~15% and ~7% for Σ_H and Σ_P , respectively, when $\Sigma_{iP} \approx \Sigma_{iS}$ ($i = H$ or P). In this way one may employ Mehta's formula (1979) for solar produced conductivities (or other such formula), and expressions for the particle produced conductivities, to find approximate values of the total conductivity.

Harel et al. [1978] formulated some very simple expressions for both Σ_P and the ratio Σ_H/Σ_P . Specifically, they find $\Sigma_P = 5.2[\text{energy flux in ergs cm}^{-2} \text{ s}^{-1}]^{0.5}$ mhos and $\Sigma_H/\Sigma_P = 0.55[\text{average particle energy in keV}]^{0.6}$. We have tested these formulas by using both monoenergetic and Maxwellian isotropic particle spectra. Results are shown in Figures 5 and 6. It is seen that the Harel et al. expression for K_p is reasonable over the range $1 \leq \bar{E} \leq 10$ keV where $\bar{E}$ is the average particle energy. It would perhaps be better, however, to include the dependence upon average particle energy in some way. Again, the Σ_H/Σ_P ratio is reasonable, but our results indicate $\Sigma_H/\Sigma_P = 0.61 \bar{E}^{5/8}$ is a slightly better fit within the range $0.5 \leq \bar{E} \leq 32$ keV.

In conclusion, we have computed conductivity distributions resulting from electron precipitation for latitudes below, within, and poleward of the auroral oval. These distributions show considerable spatial structure. To the extent that

the instantaneous electron precipitation pattern reflects the average distributions employed here, these structures must have importance in the coupling of the ionosphere and magnetosphere. We hope to explore this problem in the future.

Appendix 1: Computational Details

Ionization Profiles

Rees [1963, 1969] expresses the ionization rate per unit volume, q , per unit incident electron flux $F(\epsilon_0)$ as

$$\frac{q(\epsilon_0, z)}{F(\epsilon_0)} = \frac{\epsilon_0/\epsilon_{\text{ion}}}{\Delta\epsilon_{\text{ion}}} \lambda(z/R) \frac{\sum_i \eta_i [X_i]_z}{\sum_i \eta_i [X_i]_R} \quad (\text{A1})$$

where

- ϵ_0 the initial electron energy in eV;
- r_0 the range (atm-cm) of the electrons at the top of the atmosphere;
- $\Delta\epsilon_{\text{ion}}$ the mean energy loss per ion pair formed (35 eV);
- $\lambda(z/R)$ an energy dissipation function evaluated at partial penetration depth z/R where z is the depth in gram cm⁻² to which the electron has penetrated. z depends upon the inclination of the magnetic field

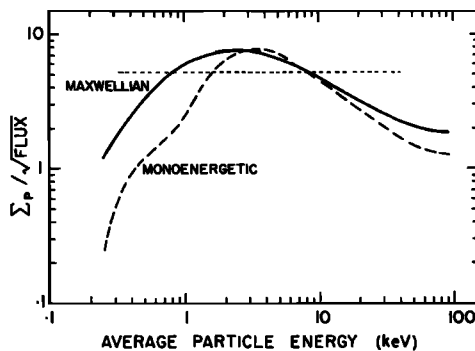


Fig. 5. Plot of the height-integrated Pedersen conductivity (mhos) per unit square root energy flux (ergs cm⁻² s⁻¹) as a function of average electron energy in keV. Monoenergetic (dashed) and Maxwellian spectra (solid) were employed. Isotropy in pitch angle was assumed. Computations were performed for 0600 UT of day 350 at 60°N and 0°E with $A_p = 4.0$, $F_{10.7} = F_{10.7} = 150$. The exospheric temperature was 846°K. The horizontal dotted line represents an approximation of Harel et al. [1978]; see text for details.

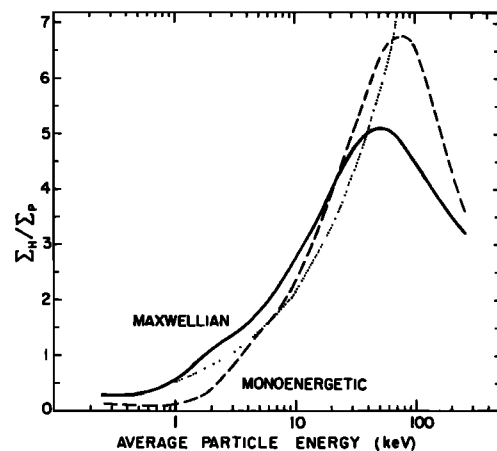


Fig. 6. Plot of the ratio of the height-integrated conductivities Σ_H/Σ_P as a function of average electron energy computed for monoenergetic and Maxwellian spectra. Computational details are the same as for Figure 5. The dotted curve represents an approximation of Harel et al. [1978]; see text for details.

[Rees, 1964]. R is the total penetration depth (gm cm^{-2}). λ is dependent upon pitch angle;

$[X_1]$ the concentrations (cm^{-3}) of neutral atmospheric constituents at depths z and R ; and

η_i relative ionization efficiencies of the atmospheric species. The summation is over all species.

In brief, one evaluates q at a set of fixed heights (z) for each incident particle energy ϵ_0 and sums (for each height) over the energy spectrum.

In the present work the above quantities have been obtained from a number of sources. $F(\epsilon_0)$ was obtained from EPD measurements as explained previously. The particle penetration ρ is the atmospheric mass density at the height of maximum penetration. The energy dissipation function λ was obtained from Berger et al. [1969] for an isotropic pitch angle distribution. For particles with initial energies below 1 keV, λ was modified to approximate the changing shape of this function as reported by Barrett and Hays [1976]. Like Jones [1975], the relative ionization efficiencies η_i were assumed to be proportional to constituent mass.

When computing the 3914-Å volume emission rate the $\eta_i[X_1]z$ term is replaced with a term $\gamma_{n2}[N_2]$

z , where γ is the fraction of N_2 ionizations that lead to 3914-Å photons. Here γ is taken as 0.054. Height integrations for the 3914-Å intensities were taken to 300 km.

As explained previously, Watt et al. [1974] height profile of effective recombination rate $\alpha(z)$ is used to compute the electron density profile as a function of altitude. At each height, z , the electron density, $N(z)$, is evaluated as

$$N(z) = \left[\left\{ \sum_{\epsilon_0} q(\epsilon_0, z) + \beta^2(\chi) P_S(z) + P_B \right\} / \alpha(z) \right]^{\frac{1}{2}} \quad (A2)$$

where $q(\epsilon_0, z)$ is the particle produced ionization resulting from the Rees computation, P_S and P_B are the solar and background production rates described in the text, β is the solar zenith angle scaling factor again described in the text, and $\alpha(z)$ is the effective recombination rate.

Conductivities

The local Pedersen and Hall conductivities are calculated from

$$\sigma_P = \frac{Ne}{B} \left[\frac{v_e \omega_e}{\omega_e^2 + \nu_e^2} + \frac{v_i \omega_i}{\omega_i^2 + \nu_i^2} \right] \quad (A3)$$

$$\sigma_H = \frac{Ne}{B} \left[\frac{\omega_e^2}{\omega_e^2 + \nu_e^2} - \frac{\omega_i^2}{\omega_i^2 + \nu_i^2} \right] \quad (A4)$$

TABLE A1

LOSS CONE PARTICLE INTENSITIES

0 < KP < 3

E = 0.15 KEV													E = 1.3 KEV												
01	03	05	07	09	11	13	15	17	19	21	23	MLT	01	03	05	07	09	11	13	15	17	19	21	23	84
84	8646	1027	1077	8736	6976	1527	6666	6756	9636	1437	5716	1077	8015	1536	4865	4955	1496	1426	1576	7205	4215	1256	9725	7245	84
83	2606	7516	1037	2037	1827	9316	4547	1097	8236	1097	9906	1577	3255	5225	5675	4725	3666	5585	7206	1696	1386	1406	3036	3066	83
82	3556	7296	7786	6437	1767	7876	2687	1477	1287	4956	1837	1087	4545	4335	4545	2686	1676	1696	1876	2706	1016	6395	4236	6215	82
81	4166	3557	4177	2757	6307	2647	3747	3127	2267	1427	6216	4996	4635	3336	1626	2426	8325	6935	2366	2676	1776	2796	4065	4545	81
80	3256	9586	3647	9547	6757	3267	1797	2637	1747	2937	6306	4686	3215	7205	1756	5046	1356	1026	1086	3516	1816	4636	5315	8415	80
79	3336	6366	2487	3017	7337	2727	3187	5267	1617	2537	1567	3826	3885	4335	1676	5356	6886	1336	9995	8145	1246	2726	1186	6075	79
78	2167	7697	2897	1088	7247	6217	1188	1668	5047	2297	1047	2956	5355	1646	3156	7696	6616	2536	5356	7206	4296	4686	1106	5405	78
77	8506	4167	8507	1008	8287	3917	1288	2898	1298	2897	7066	2686	1206	2176	3586	1167	1067	5716	6666	9856	1027	5136	1666	3455	77
76	5266	2837	3497	1468	7027	2217	1658	3148	2158	6707	2027	4726	5045	1966	2566	7476	1017	4636	1317	2007	1267	4726	2076	6885	76
75	3157	8737	1398	9137	7747	4207	1358	7338	1868	7877	4007	1557	5626	5176	1147	1087	1427	5766	4996	3927	3127	7966	9136	1786	75
74	1327	1038	8377	1138	9497	2337	3897	2468	2438	6127	4057	2407	1966	8666	6576	1297	1307	5356	2776	9006	2577	9586	1397	5986	74
73	3247	1458	1788	1018	9277	1517	1157	1958	3108	9587	4727	4147	4596	9366	1467	1337	1427	4486	2126	1437	3397	1357	1377	1117	73
72	6487	2118	1698	1168	4257	1257	1157	4637	1158	1148	8647	4507	9726	1437	1707	1997	8416	3466	2156	4126	1877	1977	1617	1157	72
71	1238	1998	1788	1228	4157	1077	1277	3187	1298	1088	7607	5587	1317	2137	1987	1897	7656	2546	2396	2616	1217	2037	1897	1217	71
70	1308	1978	1558	8327	2097	7426	8646	8016	4817	9187	1038	6707	1697	2447	2327	1637	4726	1746	1666	1216	6886	1577	2877	1457	70
69	1568	2308	2208	6577	1277	5716	7336	5086	1347	4547	1228	8917	2267	3047	2777	1377	2846	1306	1276	7605	1796	9406	2707	1807	69
68	1768	3128	2108	5087	8146	5806	6756	1417	1318	2337	7967	6397	2367	3717	2707	1117	1866	1116	9365	2606	6395	4686	1957	1657	68
67	1408	2848	1278	2997	6666	5176	5806	6076	4027	1457	3817	6347	2617	4317	2527	6796	1206	8415	5805	6575	1436	3326	9996	1617	67
66	1088	2498	1078	1777	4996	5496	5716	4216	2866	6886	2657	4337	2037	3367	1627	3396	8505	6975	5715	3765	3565	1696	7296	8776	66
65	9007	1798	7337	9136	4356	5046	6526	3956	2976	5766	9186	1337	1437	2507	1137	1616	7245	6885	4545	3785	2985	1346	2026	6756	65
64	3107	9547	3797	9046	4396	4596	5856	4236	3016	3066	4126	1267	5856	1597	5266	1296	9186	7065	4325	3285	2615	4285	7785	2756	64
63	1337	8107	2337	4116	4206	4596	6756	4186	2886	3846	2906	9006	2076	9186	2496	4545	7245	7335	4595	3345	2405	4725	4595	1946	63
62	7426	4047	1177	4596	4276	4416	6306	3966	2816	2546	2926	4906	1016	5626	1206	5535	7295	5355	4635	2955	2415	2575	5135	8555	62
61	3116	1467	6076	3256	4466	4266	6396	4356	2746	2876	2556	2166	3705	1576	6255	3645	6795	5805	4395	2995	1865	2185	4305	4815	61
60	1836	5356	3736	3586	4416	4686	7296	4296	3086	2966	2396	1916	1495	4445	2785	3625	6935	6035	4815	3195	2555	2515	3185	3735	60

E = 9.7 KEV													E > 22. KEV												
01	03	05	07	09	11	13	15	17	19	21	23	MLT	01	03	05	07	09	11	13	15	17	19	21	23	84
84	2114	1994	2194	1984	2054	1994	1934	2104	1784	6124	1964	1914	1193	1213	1263	1413	1283	1243	1133	1223	1213	1093	1193	1243	84
83	1644	1784	2014	2174	2564	2164	2434	2594	2104	2084	1714	1854	1133	1183	1263	1283	1063	1213	1463	1093	9852	1193	1143	1133	83
82	1864	1794	1814	2144	2244	2304	2554	2294	2144	1934	1994	1574	1343	1293	1303	1273	1083	1063	1313	1423	1313	1223	1483	1233	82
81	1994	2274	2164	2524	2994	2254	2454	2134	2004	1934	1724	1644	1353	1293	1273	1123	1423	1383	1593	1343	1373	1423	1143	1173	81
80	1864	2404	2524	2994	3224	2804	2074	2124	2344	2224	1344	1754	1333	1253	1243	1263	1713	1383	1333	1183	1423	1183	1233	1073	80
79	1564	2004	2534	3294	3924	4504	2464	2004	2874	1974	1844	1444	1363	1323	1293	1563	1763	1653	1323	1133	1443	1353	1293	1253	79
78	1548	1754	2784	5044	5264	6754	4684	2284	2944	2654	1954	1244	1323	1323	1153	1823	1693	2433	1743	1223	1493	1383	1463	1053	78
77	1714	1994	4274	8144	9494	1085	6884	2834	9584	3554	1694	1264	1223	1133	1563	1983	2643	2833	1713	1463	1343	1143	1363	1253	77
76	1834	2794	5804	1815	4105	2215	1265	3374	6304	1215	3384	3274	1213	1483	1733	2903	4003	5123	2123	1503	1353	3993	1183	1363	76
75	3354	2644	6844	2805	7565	5495	3255	1505	1045	9224	2295	1015	1343	1413	1443	4203	4774	2844	3273	1823	1513	1743	1253	1253	75
74	8914	3284	9944	5675	1006	9545	5225	1175	8284	2655	5495	7784	1243	1353	2113	7253	4484	5294	9273	3733	1993	1613	1643	1343	74
73	3464	4904	1865	1056	1216	1596	7695	1815	4135	2725	8595	3755	1593	1923	2263	2154	5624	1075	2954	3453	1843	1493	1623	1463	73
72	5764	1015	3535	1616	1566	1436	6525	3105	1535	6755	1716	2056	1513	2133	3573	6054	8764	2225	1914	6443	2513	2393	2043	1603	72
71	4245	2055	4995	2136	1456	1366	6935	3505	1665	5265	1406	2826	7553	4703	5883	7264	1675	2295	3544	1234	6173	3533	1963	2133	71
70	3375	4545	1066	3016	1376	1016	5675	3465	1465	4815	2476	7925	2373	9443	5034	1725	8844	1145	8264	3424	1484	6693	4253	3653	70
69	5495	1146	2516	3996	1256	7155	4725	2935	1105	5225	2366	1746	5633	1614	5874	8204	5804	1155	6324	3204	8523	6283	7283	6663	69
68	1106	2086	3576	2796	1026	5495	3655	2315	1065	5225	2026	3946	1264	3164	9914	7294	4654	1235	3954	2334	1144	8593	8953	1814	68
67	1706	3346	4016	2486	5175	3865	2705	1775	1015	3575	1476	1966	2744	1984	9814	4074	4034	1764	3844	1994	1144	9723	1634	1574	67
66	1996	4396	3826	1506	2795	3005	1845	1485	1165	9134	1356	1326	1734	1924	8054	2964	2454	2964	2324	1824	1784	2034	1614	2014	66
65	2286	4776	1586	6885	1985	1015	1715	1915	8504	7514	7385	1366	2304	1314	3274	2794	2834	2794	1954	1874	2364	2504	1614	1724	65
64	1056	2526	6935	4635	1305	1175	9494	8734	1015	8414	1715	6525	1744	1694	2944	2304	2754	2804	2374	1794	2324	2404	2054	1524	64
63	5135	1316	3685	8054	8234	6664	8824	7064	5714	4164	1715	4325	2014	1694	1964	2394	3254	2094	1934	1624	2374	2574	2584	1444	63
62	2855	6845	1835	6344	6344	5444	6754	5084	4724	2924	6484	8594	1604	1314	2534	2084	2524	1944	1684	1634	2774	2344	2144	1664	62
61	3454	1975	7424	4904	5404	4634	6844	4684	3434	3604	2654	2474	1084	1374	2524	1784	2184	1424	1214	3844	2494	1684	1574	1674	61
60	1784	4484	3664	4124	5044	4464	7564	5954	3324	3574	2464	1634	7273	8963	1734	1324	1544	8993	5613	9893	3414	1054	9163	1644	60

where all quantities except e , the electronic charge, are functions of height; ω and ν are gyro and collision frequencies, respectively; the subscripts e and i indicating electrons or ions. Brekke et al.'s [1972] expression for the ion collision frequency $\nu_i = 7.5 \times 10^{-10} N_N$, where N_N is the neutral number density (cm^{-3}), was used. The electron collision frequency is taken from Hanson [1965].

$$\nu_e = 5.4 \times 10^{-10} N_N T_e^{\frac{1}{2}} + N\{34 + 4.18 \log_{10}(T_e^3/N)\} T_e^{-3/2} \quad (\text{A5})$$

In order to evaluate this latter expression we have employed a computed electron temperature under auroral conditions [Walker and Rees, 1968] as being representative. Results are not sensitive to this choice as it is only at low altitudes where $\nu_e \approx \omega_e$ where the electron temperature becomes important, and there the electron temperature is closely tied to the neutral temperature, while N is small and B is large. In computing the ion gyro frequency the mean molecular mass was used. The intensity of the earth's magnetic field B was computed from the IGRF 1975 model field at each grid point and altitude z .

The computational procedure then requires for each grid point fitting of power law spectra to

the input particle data, the evaluation of the particle fluxes at 33 energies and the computation of the particle ionization rate $\sum_{\epsilon_0} q(\epsilon_0, z)$ for a number of altitudes, z . Nineteen, nonuniformly spaced altitudes between 70 and 220 km are used. The electron density is evaluated for each z by (A2). Then B , ω_e , ω_i , ν_i , ν_e are evaluated at each height. Finally, (A3) and (A4) are employed to compute the local conductivities. Height integrals are carried out by fitting exponential functions to successive pairs of altitude points, z_1 and z_2 , and computing the integral of these exponentials from z_1 to z_2 . The sum of these integrals is then the desired conductivities.

Appendix 2

We tabulate here, Tables A1 and A2, the particle intensities at 0.15, 1.27, and 9.65 keV (differential intensities $\text{cm}^{-2} \text{s}^{-1} \text{sr}^{-1} \text{keV}^{-1}$) and >22 keV (integral intensity $\text{cm}^{-2} \text{s}^{-1} \text{sr}^{-1}$) used as inputs in this paper. Four digits are given, the last being the power of 10. An implicit decimal point occurs between the first two digits. Thus 1234 should be read as 1.23×10^4 . For the low K_p case (Table A1) approximately 650 samples were acquired per bin from ~150 passes of Isis 2. The higher K_p table is based upon ~230 samples per bin acquired from 45 passes.

Tables A3-A7 present the computed values of the

TABLE A2
LOSS CONE PARTICLE INTENSITIES

3 < KP < 9																										
E = 0.15 KEV												E = 1.3 KEV														
01	03	05	07	09	11	13	15	17	19	21	23	MLT	01	03	05	07	09	11	13	15	17	19	21	23		
84	1536	1916	1766	3926	1876	1986	2496	1706	1806	5406	1746	1396	1165	1405	1145	3775	1755	1265	1635	1415	1425	7875	1285	9994	84	
83	9455	1816	2456	3316	2006	2336	2376	1826	2176	2996	2116	1126	8684	1215	1835	5135	1605	1675	1515	1315	1445	1515	1915	9634	83	
82	1476	8106	4087	1147	1966	2626	1856	2126	4386	1866	7206	1136	9094	2885	7565	5045	1725	1665	1395	1565	7695	1485	1056	9274	82	
81	1676	3946	2846	7566	1457	9096	1836	2836	1876	1826	1426	1406	1795	3165	2415	6435	1326	1456	1205	1905	1175	1885	1125	1035	81	
80	1866	5086	1467	3037	1037	2527	2807	3086	1936	1896	2366	4276	2245	4595	5945	2256	1136	2886	2615	2205	1375	1855	2215	1405	80	
79	7336	4266	1457	1827	2137	1367	7336	4136	1387	2356	4546	3936	8555	4225	1286	1686	2526	1586	8825	2605	6935	1975	6755	2185	79	
78	3716	1257	5627	5897	5987	2437	5407	5046	1157	2426	5536	6706	4865	1866	2966	2356	6796	3665	1256	2735	7115	2675	8105	6485	78	
77	7606	9226	2657	3987	6937	2477	2108	1328	1197	1117	3516	5536	1056	1096	2156	3056	7746	5045	1626	2166	4636	2646	2245	8555	77	
76	2617	4507	1348	1158	7657	1848	5587	3667	1018	3847	2976	2007	4636	1486	8236	1457	1097	4636	1366	2266	1927	3816	3065	2196	76	
75	1698	1088	7567	4427	2668	1317	6797	2608	1248	3587	1997	1087	3866	1677	7656	5176	2767	1246	1337	2657	2387	4686	3306	1646	75	
74	4367	2408	8827	2888	5087	1177	8057	5677	1248	5317	7877	2287	3596	2387	8467	744	6436	1086	3036	7836	2947	6976	1407	6166	74	
73	6037	1508	1038	1268	6077	1037	1397	4278	2238	1158	3117	1837	9766	2147	1057	1237	1047	2006	1216	1667	2317	1647	6976	2876	73	
72	2128	9407	8237	1408	6528	6796	1867	3158	3418	1528	5177	3697	1317	7966	1537	1407	4127	1576	1216	1367	3787	2127	1267	8196	72	
71	2058	2748	1288	1458	7517	1397	1597	4508	2578	1198	4317	2837	1357	4507	1967	2067	1577	3676	3196	4377	3997	2747	9906	5856	71	
70	9497	1398	2308	2838	5317	1227	7246	3777	4908	2078	1268	3607	1647	2457	2577	2107	9636	2836	8055	2327	6037	2387	2407	7476	70	
69	1208	1128	7608	2168	3227	5586	6076	4386	1468	1858	1538	3627	9226	3687	3487	2987	7966	1486	1126	7066	4377	2927	3527	7606	69	
68	1078	2478	2778	2168	3227	5586	6076	4386	1468	1858	1538	3627	1547	1627	6977	3467	8956	1186	7835	5225	1847	3287	3887	7656	68	
67	8327	2988	3028	1508	1667	5046	6756	2996	2507	1018	1278	3857	1417	2297	4487	3997	7246	6665	6845	1697	2627	3387	8596	67	67	
66	1078	2318	2988	9727	1137	7206	7066	3786	9766	1988	7657	4997	1157	3837	5177	3317	3716	5225	6435	3025	2736	1417	2617	9996	66	
65	8107	2948	4298	6307	6796	7026	6756	3806	2896	3567	5407	5447	1177	5407	3677	8466	9446	5675	5625	5675	2995	1775	4956	1387	1127	65
64	3547	3488	1718	2817	5496	4816	7386	3716	2436	9547	4687	3487	8286	3607	2387	3046	5445	4415	5535	3625	1855	7066	1117	7066	63	
63	4807	1478	1428	2607	4866	7296	7516	3826	2746	7066	2207	2387	6436	2107	1577	1696	4445	5625	5175	2935	2125	1276	4026	4726	62	
62	3977	5897	5857	6666	4496	7066	8866	4406	3096	3196	3296	1797	6076	8866	7026	1216	3805	3965	5715	2895	2135	2895	3395	3746	61	
61	2887	5767	3377	4066	4136	5676	9946	3786	2876	2996	1806	7836	3986	5586	2976	4725	3395	4995	7295	2805	1655	2095	1235	1546	60	

E = 9.7 KEV												E > 22. KEV													
01	03	05	07	09	11	13	15	17	19	21	23	MLT	01	03	05	07	09	11	13	15	17	19	21	23	
84	1504	1444	1634	1784	1754	1784	1894	1494	1804	1654	1354	1244	1143	1513	1303	9732	1433	1353	1243	1433	1323	1243	1193	1473	84
83	1244	1624	1874	1794	2094	2064	1894	1844	1674	1684	1264	1244	8412	1183	1263	1313	1913	1213	1123	1413	1563	1673	1203	1603	83
82	1254	1594	2314	1934	1864	1894	1964	1854	1594	1674	1374	1284	1403	7572	1263	1243	1323	1643	1393	1653	1673	1313	1373	1283	82
81	1794	2064	1994	1764	1714	1534	1514	1664	1844	1984	1244	1244	1253	1463	1133	7972	1493	1683	1493	1013	1333	1563	1123	1943	81
80	2694	5174	2524	2514	2024	3064	1524	1974	1624	1944	1894	2294	7142	1093	1133	1883	1673	1443	7412	1063	1063	2253	1743	8332	80
79	2394	3914	2474	6034	3214	2974	1934	1714	1694	1874	1894	1975	1603	1483	1153	1633	1473	1623	1463	8072	1593	1773	1423	1143	79
78	3104	4304	2884	7924	1215	2434	2294	2054	1724	2054	2456	4724	1363	1033	9432	1353	1813	1303	1463	1223	7822	1253	1883	8382	78
77	4994	4344	6704	1945	1775	1125	2254	2134	2114	2014	3164	3704	5072	7532	1423	1333	2273	1643	8252	1813	1513	1833	1303	1143	77
76	3115	1155	1215	1785	3065	9044	2474	2134	2924	2284	3554	1945	4004	1443	1703	4023	3593	2663	1333	1553	6242	1043	8262	1143	76
75	2205	2155	3545	2195	3895	1965	2724	2744	7384	4634	7154	5405	1063	2783	1073	1733	5233	3553	1343	1873	1583	2053	1243	2023	75
74	7155	5855	1505	1556	6255	6165	1195	2486	3485	3075	2515	6755	8212	3023	1353	5593	3584	1254	1933	1563	1073	1023	1633	1013	74
73	5265	3285	3555	1546	1116	9225	1895	3754	1095	9815	2595	7515	2653	2533	2493	1044	3973	2554	1873	1903	1593	1793	1893	9563	73
72	5805	3325	4725	1726	1846	8105	3625	4234	2635	2816	2505	1606	5024	6923	4243	1224	3655	1925	6133	3053	3093	1993	1913	7973	72
71	8055	3895	6395	2156	3276	1196	3635	9314	1246	3275	1286	1496	1784	2913	4153	3164	3935	9714	4223	3203	1266	1363	2693	1173	71
70	5985	8645	1836	3176	2386	1746	3855	1735	6615	3465	1956	1906	1474	5433	2705	2984	1386	3435	3654	4413	9503	1763	3153	1274	70
69	8645	1206	3936	4326	2466	1576	3655	1585	9585	6165	9316	4476	2394	3374	2695	1515	2125	1426	4594	5363	3504	4253	1444	9762	69
68	1356	1486	5446	6256	2906	1726	4175	7924	9455	6526	8736	2546	6484	1324	1835	7924	7015	3556	1155	4893	4703	8913	6244	3075	68
67	1696	2236	9856	8236	3366	7785	4505	1235	4995	2176	1296	4996	6164	4704	8355	2565	5845	1694	1055	1084	2873	2724	3214	6255	67
66	2966	4466	1047	7926	3056	3505	3375	1985	6934	1506	6396	4106	2965	3265	9775	4116	4735	3994	1125	1434	6103	1754	3284	4895	66
65	3826	8916	1017	4426	2056	4995	2955	1745	5174	2166	3076	4506	5105	2985	6166	4946	1275	1905	1675	1754	1634	1964	3904	7665	65
64	5406	8776	7336	2966	7515	3605	2115	1445	7654	5315	2556	2516	5004	2275	1266	3435	1095	3115	1175	2044	4314	2584	2644	5404	64
63	4956	7786	4506	7745	4905	1315	2655	7744	4724	4545	1966	1396	3544	1295	4925	2475	1805	6404	8664	1744	1754	2624	3074	2864	63
62	5046	3246	2386	5805	2495	1625	1055	5894	6934	1855	1605	1386	2374	2324	6004	2384	1615	9844	6994	1494	1894	4904	4744	2934	62
61	3416	1486	9005	1545	1345	7874	9634	4954	3314	5444	3064	6305	1854	2784	6474	2984	3344	2084	7994	2784	1884	4844	4404	3964	61
60	1246	6845	1775	7204	6884	914	1054	4684	3254	3824	2114	3285	1704	2214	1414	9234	2254	1234	2434	2844	1334	3714	2434	3474	60

TABLE A3
SOLAR AND BACKGROUND
DAY 91 UT 1700

PEDERSEN CONDUCTIVITY													UNITS (0.1 MHO)					HALL CONDUCTIVITY												
1	3	5	7	9	11	13	15	17	19	21	23	MLT	1	3	5	7	9	11	13	15	17	19	21	23						
84	37	39	43	47	52	55	56	55	51	46	41	38	65	69	76	84	91	96	98	96	89	80	72	66	84					
83	35	38	42	48	53	56	58	56	52	46	40	36	62	67	74	84	92	99	101	98	90	80	70	63	83					
82	34	37	42	48	53	57	59	57	52	45	39	34	59	64	74	84	94	101	103	100	91	79	68	60	82					
81	32	35	41	48	54	59	61	59	53	45	38	33	57	62	73	84	95	103	106	103	93	79	66	58	81					
80	30	34	41	48	55	60	62	60	54	45	36	31	54	60	72	85	97	105	109	105	94	78	64	55	80					
79	29	33	40	48	56	61	64	62	55	45	35	30	51	58	71	85	98	107	111	107	95	78	62	52	79					
78	27	32	40	48	56	63	65	63	55	44	34	28	48	56	70	85	99	110	114	110	96	77	60	49	78					
77	26	30	39	49	57	64	67	65	56	44	33	26	45	54	69	86	101	112	117	112	97	77	58	46	77					
76	24	29	39	49	58	65	68	66	57	44	32	25	42	52	68	86	102	114	119	114	98	76	55	43	76					
75	22	28	38	49	59	66	70	67	58	44	31	23	39	50	67	86	103	116	122	117	100	76	53	40	75					
74	21	27	38	49	60	68	72	69	58	43	29	21	37	48	66	87	105	118	125	119	101	75	51	37	74					
73	19	26	37	49	60	69	73	70	59	43	28	20	34	46	66	87	106	120	127	122	102	74	49	34	73					
72	17	25	37	50	61	70	75	72	60	43	27	18	31	44	65	88	108	123	130	124	103	74	47	32	72					
71	16	23	36	50	62	71	76	74	61	42	26	16	28	42	64	88	109	125	133	127	104	73	45	29	71					
70	14	22	36	50	63	73	78	75	61	42	24	15	25	40	63	89	111	127	136	129	105	72	42	26	70					
69	13	21	36	51	64	74	80	77	62	41	23	13	23	38	63	89	112	130	138	132	107	71	40	23	69					
68	11	20	35	51	65	76	82	78	63	41	22	11	20	36	62	90	114	132	141	135	108	71	38	20	68					
67	10	19	35	52	66	77	83	80	64	41	21	10	17	34	62	91	115	135	144	137	109	70	36	17	67					
66	8	18	35	52	67	78	85	82	64	40	19	8	14	32	61	91	117	137	147	140	110	69	34	15	66					
65	7	17	34	52	68	80	87	83	65	40	18	7	12	30	60	92	119	140	150	143	111	68	31	12	65					
64	5	16	34	53	69	82	89	85	66	39	17	5	9	28	60	93	121	142	154	145	112	67	29	9	64					
63	4	14	34	53	70	83	91	87	67	39	15	4	7	26	59	94	123	145	157	148	113	67	27	7	63					
62	2	13	33	54	71	85	93	89	67	38	14	2	4	24	59	95	125	148	160	151	115	66	25	4	62					
61	1	12	33	55	73	87	95	91	68	38	13	1	2	22	58	96	127	151	164	154	116	65	23	2	61					
60	1	11	33	55	74	89	97	93	69	37	12	1	2	20	58	97	129	154	167	157	117	64	20	2	60					

TABLE A4
PARTICLES AND BACKGROUND

DAY 91													UT 1700				0 < KP < 3												
PEDERSEN CONDUCTIVITY													UNITS (0.1 MHO)				HALL CONDUCTIVITY												
1	3	5	7	9	11	13	15	17	19	21	23	MLT	1	3	5	7	9	11	13	15	17	19	21	23					
84	7	8	6	6	8	8	8	7	5	12	7	6	8	8	8	7	8	8	8	8	6	16	8	7	84				
83	4	5	6	6	12	6	15	10	8	9	10	10	5	6	7	7	11	8	13	10	8	9	8	9	83				
82	5	5	5	10	9	9	10	11	8	6	12	5	7	6	6	10	9	9	10	11	9	8	11	6	82				
81	6	11	9	10	8	7	10	11	9	11	5	5	7	10	9	10	10	8	11	10	9	10	6	6	81				
80	5	7	9	14	9	8	8	12	10	13	4	6	6	9	10	13	11	9	8	10	11	12	4	7	80				
79	4	5	9	14	16	10	8	7	10	11	8	5	5	7	10	14	15	13	9	8	12	10	8	5	79				
78	5	8	12	18	17	14	16	15	15	14	8	3	5	7	11	18	17	18	17	12	14	14	9	2	78				
77	7	9	14	23	22	20	19	18	26	16	8	2	7	9	15	23	24	23	20	15	27	16	8	2	77				
76	5	10	14	24	32	23	27	25	22	12	8		6	11	16	30	41	31	29	19	24	27	13	11	76				
75	15	14	22	30	41	31	27	42	38	24	30	15	14	12	21	37	60	49	37	38	34	27	36	20	75				
74	15	17	21	37	43	37	28	26	34	33	42	19	19	15	23	50	65	61	44	29	31	40	53	21	74				
73	14	20	30	45	47	42	30	33	53	37	47	34	14	19	33	66	71	76	54	37	57	43	62	42	73				
72	20	26	37	56	46	40	29	28	36	51	59	55	20	27	44	84	77	77	49	40	37	64	83	80	72				
71	36	35	42	60	44	37	30	26	33	49	58	61	43	38	51	93	78	75	53	42	38	60	78	92	71				
70	36	44	54	65	40	31	27	23	27	45	75	44	42	52	76	110	70	61	52	43	36	58	104	58	70				
69	43	58	71	63	35	27	24	20	18	41	73	57	53	76	106	105	63	53	46	39	27	56	103	81	69				
68	52	72	79	59	31	24	21	24	14	35	65	72	70	99	124	99	56	49	40	38	25	53	94	113	68				
67	60	84	81	52	23	20	17	17	17	29	52	57	85	118	129	88	42	40	34	31	27	45	80	85	67				
66	59	87	74	40	17	17	15	14	13	17	48	45	86	127	122	68	31	33	28	27	26	27	76	70	66				
65	57	84	53	27	15	12	14	13	12	15	31	43	88	126	82	47	28	22	27	26	24	26	54	68	65				
64	38	64	36	23	13	12	11	11	12	12	16	29	60	96	56	42	24	23	22	22	25	24	30	48	64				
63	25	47	25	10	11	10	11	11	10	10	15	23	42	72	42	20	21	18	21	21	21	19	29	39	63				
62	18	35	18	10	10	9	10	9	9	7	11	12	32	54	31	19	18	16	19	18	20	15	20	20	62				
61	7	19	12	8	9	9	10	9	7	7	7	6	12	31	22	16	17	15	18	17	16	16	13	11	61				
60	3	9	7	8	9	8	10	9	8	8	6	4	7	15	15	14	16	14	17	16	18	15	11	8	60				

TABLE A5

PARTICLES AND BACKGROUND

DAY 91 UT 1700 3 < KP < 9

	PEDERSEN CONDUCTIVITY												UNITS (0.1 MHO)					HALL CONDUCTIVITY											
	1	3	5	7	9	11	13	15	17	19	21	23	MLT	1	3	5	7	9	11	13	15	17	19	21	23				
84	2	2	2	5	3	3	4	3	3	6	2	3		4	4	4	5	5	5	6	4	6	7	3	4	84			
83	3	2	4	5	4	4	3	3	3	3	2	1		4	4	6	6	7	6	5	6	5	5	2	2	83			
82	2	4	7	5	4	4	3	4	6	3	6	3		3	4	8	7	5	6	6	6	7	5	5	3	82			
81	4	5	4	5	7	7	2	4	3	4	3	4		5	7	6	6	7	6	4	5	5	7	5	6	81			
80	5	8	7	10	7	11	4	4	3	4	4	4		7	12	8	11	8	12	4	6	4	7	7	6	80			
79	7	7	8	12	11	9	7	4	6	4	6	12		9	11	9	16	12	11	8	5	7	7	8	21	79			
78	7	11	11	14	21	6	8	5	6	5	40	9		10	13	11	18	24	8	9	7	6	7	74	12	78			
77	10	10	14	20	24	11	9	10	13	11	6	9		12	12	17	26	29	18	8	10	12	11	9	12	77			
76	27	15	23	29	30	18	9	10	24	13	7	19		43	21	26	32	37	21	10	10	17	12	9	27	76			
75	24	33	30	23	40	16	19	26	32	17	17	25		30	36	37	30	46	25	15	19	29	19	20	41	75			
74	31	46	24	53	33	24	18	58	48	32	35	35		43	55	28	74	51	42	23	88	51	39	40	47	74			
73	35	39	32	49	42	30	17	24	35	53	29	31		47	43	40	73	59	53	25	20	33	71	37	52	73			
72	40	30	38	52	67	29	21	31	50	76	34	48		58	41	48	77	104	60	35	24	51	109	39	75	72			
71	43	49	44	61	66	37	25	40	76	47	49	43		61	51	55	90	118	67	37	34	133	51	72	65	71			
70	41	51	65	68	57	41	21	38	70	46	67	50		55	64	104	104	120	86	40	40	78	51	92	80	70			
69	40	62	86	82	53	39	22	27	71	56	116	63		61	82	139	131	96	100	41	33	90	67	181	98	69			
68	51	54	109	94	58	42	22	12	57	110	117	57		78	78	162	146	116	122	46	19	76	166	186	108	68			
67	53	66	122	106	59	24	22	13	47	77	66	73		84	99	215	175	117	44	47	24	59	112	89	146	67			
66	62	92	128	108	52	18	20	15	18	61	98	69		112	149	226	226	107	36	43	29	24	91	156	133	66			
65	71	113	127	85	40	22	19	14	15	58	73	73		131	190	273	193	79	49	44	29	22	97	115	144	65			
64	73	120	111	60	26	19	17	14	10	37	65	57		124	192	203	113	52	47	37	28	24	58	103	94	64			
63	67	107	87	33	21	12	18	10	8	38	58	48		114	173	154	66	47	26	38	21	18	57	92	79	63			
62	65	73	65	25	16	14	13	10	10	21	23	40		111	111	103	44	37	30	27	20	21	38	36	67	62			
61	56	49	42	16	12	10	13	9	7	10	7	30		93	78	69	28	24	18	27	19	16	22	16	50	61			
60	37	36	22	10	9	11	13	9	7	8	4	21		60	56	32	18	18	19	24	19	14	18	11	37	60			

TABLE A6

PARTICLES, SOLAR AND BACKGROUND

DAY 91 UT 1700 0 < KP < 3

PEDERSEN CONDUCTIVITY													UNITS (0.1 MHO)				HALL CONDUCTIVITY												
1	3	5	7	9	11	13	15	17	19	21	23	MLT	1	3	5	7	9	11	13	15	17	19	21	23					
84	38	40	43	48	52	55	57	55	51	48	41	38	66	69	76	84	91	96	98	96	89	82	72	66	84				
83	36	38	43	48	54	56	60	57	52	46	41	38	62	67	75	84	93	99	102	98	90	80	70	64	83				
82	34	37	42	49	54	58	60	58	53	46	41	35	60	65	74	85	94	101	104	101	91	79	69	61	82				
81	33	37	42	49	55	59	61	60	54	46	38	33	57	63	73	85	95	103	106	103	93	79	66	58	81				
80	31	35	42	50	56	60	63	61	55	47	37	32	54	61	72	86	97	105	109	105	94	79	64	55	80				
79	29	33	41	50	58	62	64	62	55	46	36	30	51	58	71	86	99	108	111	107	95	78	62	52	79				
78	28	33	41	52	59	64	67	65	57	47	35	28	48	56	71	88	101	111	115	110	97	79	60	49	78				
77	26	32	42	54	62	67	70	67	62	47	34	26	46	55	71	90	105	115	119	113	103	79	58	46	77				
76	24	31	41	55	67	69	74	71	63	50	34	26	43	53	71	93	114	120	124	117	103	83	58	45	76				
75	27	32	45	58	73	74	76	81	70	51	44	28	43	52	72	97	125	129	130	126	108	82	69	47	75				
74	26	33	43	63	75	78	78	74	69	56	53	29	43	51	72	106	130	138	135	124	108	90	81	46	74				
73	24	33	49	69	78	83	80	78	81	58	57	40	38	51	77	117	135	150	142	129	124	91	88	60	73				
72	27	37	53	77	78	82	81	78	71	68	67	59	39	54	84	131	140	151	142	133	113	106	105	94	72				
71	40	43	57	81	78	82	83	79	70	66	65	65	57	61	89	139	141	151	146	136	114	102	100	104	71				
70	40	50	67	86	76	80	83	79	68	63	81	47	54	72	107	153	137	145	148	138	113	99	122	70	70				
69	46	63	82	84	74	80	84	80	65	60	79	60	62	92	134	149	135	143	148	139	111	97	120	90	69				
68	55	76	89	81	73	80	85	82	65	55	71	74	77	113	149	145	132	143	148	141	111	94	110	120	68				
67	62	88	91	76	70	80	85	82	66	51	58	59	90	130	154	137	125	142	149	141	113	88	95	91	67				
66	61	90	84	67	69	81	87	83	66	44	53	47	90	139	147	122	122	142	150	143	114	75	89	75	66				
65	58	87	65	60	70	81	88	84	66	43	37	44	91	138	111	108	123	141	153	145	114	74	68	72	65				
64	39	67	51	59	70	82	90	86	67	41	24	30	62	107	88	105	123	144	155	147	115	72	44	50	64				
63	26	51	43	54	71	84	92	87	67	40	22	24	44	82	76	96	124	146	158	149	115	69	42	41	63				
62	18	39	34	55	72	85	93	89	68	39	18	12	33	63	69	96	126	148	161	152	116	67	33	21	62				
61	7	23	35	55	73	87	95	91	68	39	15	6	12	40	63	97	127	151	164	154	116	67	26	12	61				
60	3	14	34	56	74	89	98	93	69	38	13	5	7	26	60	97	129	154	168	157	118	66	24	8	60				

TABLE A7
PARTICLES, SOLAR AND BACKGROUND
DAY 91 UT 1700 3 < KP < 9

PEDERSEN CONDUCTIVITY													UNITS (0.1 MHO)					HALL CONDUCTIVITY												
	1	3	5	7	9	11	13	15	17	19	21	23	MLT	1	3	5	7	9	11	13	15	17	19	21	23					
84	37	39	43	48	52	55	56	55	51	46	41	38		65	69	75	83	91	96	98	95	89	80	71	66	84				
83	35	38	42	48	53	56	58	56	52	46	40	36		62	66	74	84	92	98	100	98	90	79	69	63	83				
82	34	37	42	48	53	57	59	57	53	45	39	34		59	64	74	84	94	101	103	100	91	79	68	60	82				
81	32	36	41	48	54	59	61	59	53	45	37	33		57	63	73	84	95	103	106	102	92	79	65	57	81				
80	31	35	41	49	55	61	62	60	54	45	37	31		54	62	72	85	97	106	108	105	93	78	64	55	80				
79	30	34	41	50	57	62	64	62	55	45	36	32		52	59	71	87	99	108	111	107	95	78	62	58	79				
78	28	34	41	51	61	63	66	63	56	45	36	29		49	58	71	88	103	109	114	109	96	77	105	51	78				
77	28	32	42	53	62	65	67	65	58	45	33	28		48	56	72	92	106	113	116	112	98	77	58	48	77				
76	37	33	46	57	66	68	69	67	62	46	32	32		64	58	75	95	112	116	119	114	101	77	56	54	76				
75	34	44	49	55	73	68	73	73	67	47	35	35		54	66	82	94	118	120	123	119	106	79	59	63	75				
74	38	55	46	75	69	72	74	93	77	55	47	42		63	80	75	124	121	129	127	161	119	90	70	67	74				
73	41	48	51	72	75	76	75	75	70	71	42	38		64	69	82	123	129	136	130	124	110	112	66	68	73				
72	45	40	55	75	94	77	78	79	80	91	44	53		70	64	87	127	158	140	136	128	120	146	67	88	72				
71	47	56	59	81	94	82	81	85	100	65	57	48		72	73	92	137	171	147	140	134	175	96	93	78	71				
70	45	57	77	88	88	86	81	85	96	64	73	-53		65	83	130	150	171	160	143	138	141	95	111	90	70				
69	43	67	96	100	86	86	83	82	97	72	121	66		69	97	163	171	157	169	146	137	149	107	196	108	69				
68	53	59	118	111	90	89	85	79	87	121	121	59		85	93	185	188	171	186	150	136	141	197	199	113	68				
67	55	70	130	122	92	81	87	81	81	90	71	75		89	111	234	212	174	145	153	139	130	144	163	149	67				
66	63	95	136	124	88	81	88	83	67	75	102	70		116	158	244	254	168	143	154	143	113	124	168	136	66				
65	72	116	134	104	81	84	90	85	67	73	76	74		134	198	287	222	150	150	158	146	113	130	126	147	65				
64	74	123	119	83	74	84	91	86	67	55	69	58		126	199	220	156	135	151	158	148	115	95	114	96	64				
63	68	110	96	64	74	84	93	87	67	56	62	49		116	180	174	119	133	147	162	149	114	93	102	81	63				
62	65	76	76	61	73	86	94	89	68	44	28	40		111	119	128	108	131	151	162	152	116	78	46	68	62				
61	56	52	56	57	73	87	96	91	68	39	15	30		93	85	97	101	129	152	166	155	116	69	28	50	61				
60	37	39	40	56	74	89	98	93	69	38	12	21		61	63	69	98	130	155	168	158	117	67	23	37	60				

height-integrated conductivities Σ_p and Σ_H from which Figures 1, 2, and 4 were drawn. The reader is advised that these values are those resulting from the computation and are not smoothed in any way. These values are given in units of $1/10$ mho.

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